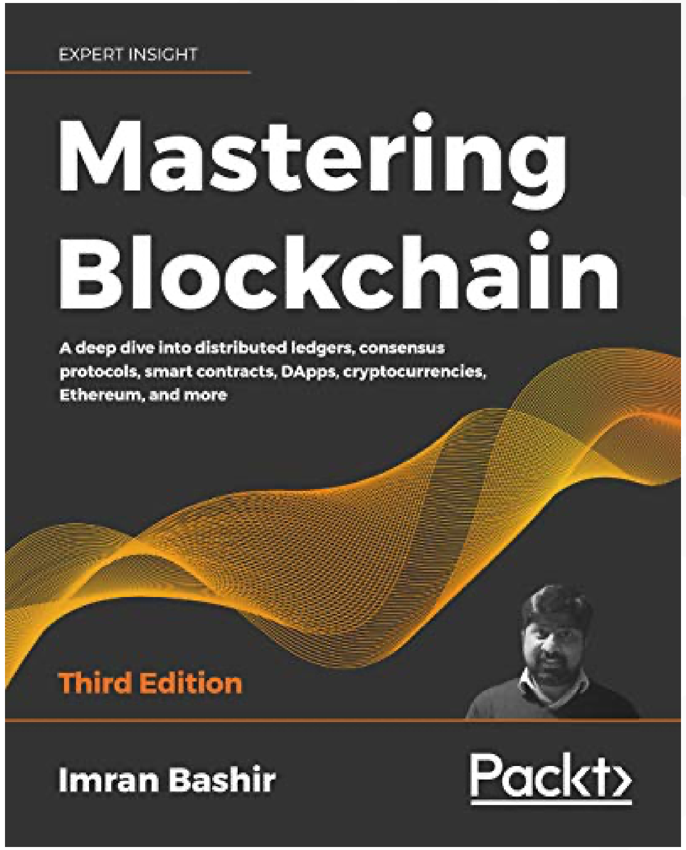
Mastering Blockchain

Third Edition

Chapter 6, Introducing Bitcoin



Outline

- Bitcoin—an overview
- Cryptographic keys
- Transactions
- Blockchain
- Mining

Bitcoin—a user's perspective

- How to send and receive payments
- Steps:
  - The transaction starts with a sender signing the transaction with their private key.
  - The transaction is serialized so that it can be transmitted over the network.
  - The transaction is broadcast to the network.
  - Miners listening for transactions pick up the transaction.
  - The transaction is verified for its legitimacy by the miners.
  - The transaction is added to the candidate/proposed block for mining.
  - Once mined, the result is broadcast to all nodes on the Bitcoin network.
  - Usually, at this point, users wait for up to six confirmations to be received before a transaction is considered final; however, a transaction can be considered final at the previous step. Confirmations serve as an additional mechanism to ensure that there is probabilistically a very low chance for a transaction to be reverted, but otherwise, once a mined block is finalized and announced, the transactions within that block are final at that point.

Cryptographic keys

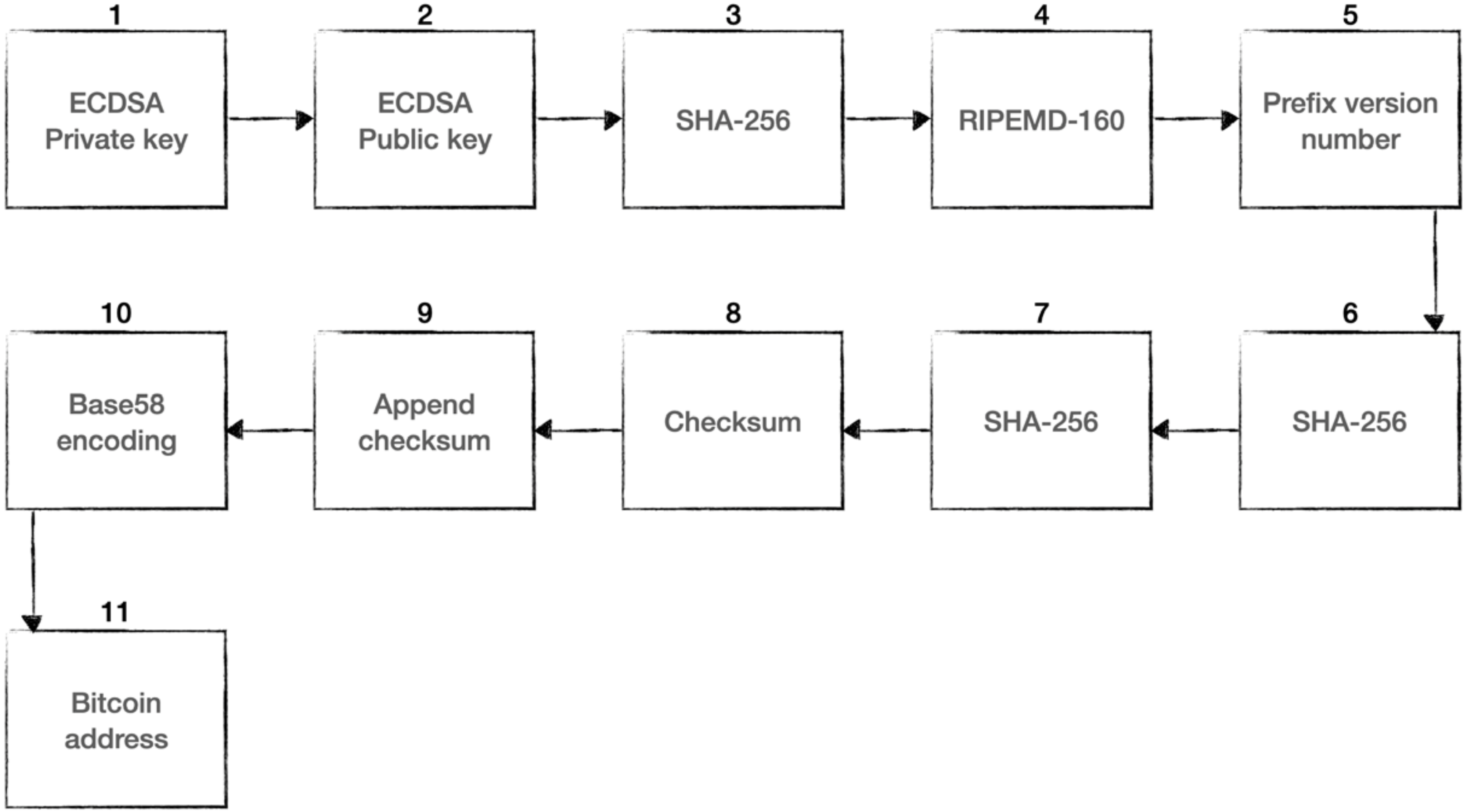
- Private keys in Bitcoin
  - Private keys are used to digitally sign the transactions, proving ownership of the bitcoins.
- Public keys in Bitcoin
  - Public keys are used by nodes to verify that the transaction has indeed been signed with the corresponding private key.
- Addresses in Bitcoin
  - A Bitcoin address is created by taking the corresponding public key of a private key and hashing it twice, first with the SHA256 algorithm and then with RIPEMD160.
- Bitcoin addresses are encoded using Base58Check encoding

QR code of the Bitcoin address
1ANAguGG8bikEv2fYsTBnRUmx7QUcK58wt

From *bitaddress.org*,
a private key and Bitcoin address
in a paper wallet

Address generation in Bitcoin

To generate an address in Bitcoin, an 11 step process is used, as depicted here:



Transactions

- A user/sender sends a transaction using wallet software or some other interface.
- The transaction is signed using the sender's private key.
- The transaction is broadcasted to the Bitcoin network using a flooding algorithm.
- Mining nodes (miners) who are listening for the transactions verify and include this transaction in the next block to be mined.
- Next, the mining starts.
- Finally, the confirmations start to appear in the receiver's wallet.

The transaction data structure

Field	Size	Description
Version number	4 bytes	Specifies the rules to be used by the miners and nodes for transaction processing. There are two versions of transactions, that is, 1 and 2.
Input counter	1–9 bytes	The number (a positive integer) of inputs included in the transaction.
List of inputs	Variable	Each input is composed of several fields. These include: • The previous transaction hash • The index of the previous transaction • Transaction script length • Transaction script • Sequence number The first transaction in a block is also called a coinbase transaction. It specifies one or more transaction inputs. In summary, this field describes which Bitcoins are going to be spent.
Output counter	1–9 bytes	A positive integer representing the number of outputs.
List of outputs	Variable	Outputs included in the transaction. This field depicts the target recipient(s) of the Bitcoins.
Lock time	4 bytes	This field defines the earliest time when a transaction becomes valid. It is either a Unix timestamp or the block height.

The transaction data structure – inputs and outputs

Transaction input data structure

Field	**Size**	**Description**
Transaction hash	32 bytes	The hash of the previous transaction with UTXO
Output index	4 bytes	This is the previous transaction's output index, such as UTXO to be spent
Script length	1-9 bytes	Size of the unlocking script
Unlocking script	Variable	Input script (`ScriptSig`), which satisfies the requirements of the locking script
Sequence number	4 bytes	Usually disabled or contains lock time — disabled is represented by `0xFFFFFFFF`

Transaction output data structure

Field	**Size**	**Description**
Value	8 bytes	The total number (in positive integers) of Satoshis to be transferred
Script size	1 – 9 bytes	Size of the locking script
Locking script	Variable	Output script (`ScriptPubKey`)

Script

- Simple stack-based language used to describe how bitcoins can be spent and transferred
- Evaluated from left to right using a Last in, First Out (LIFO) stack
- Composed of two components: elements and operations.
- Scripts use various operations (opcodes) to define their operations.

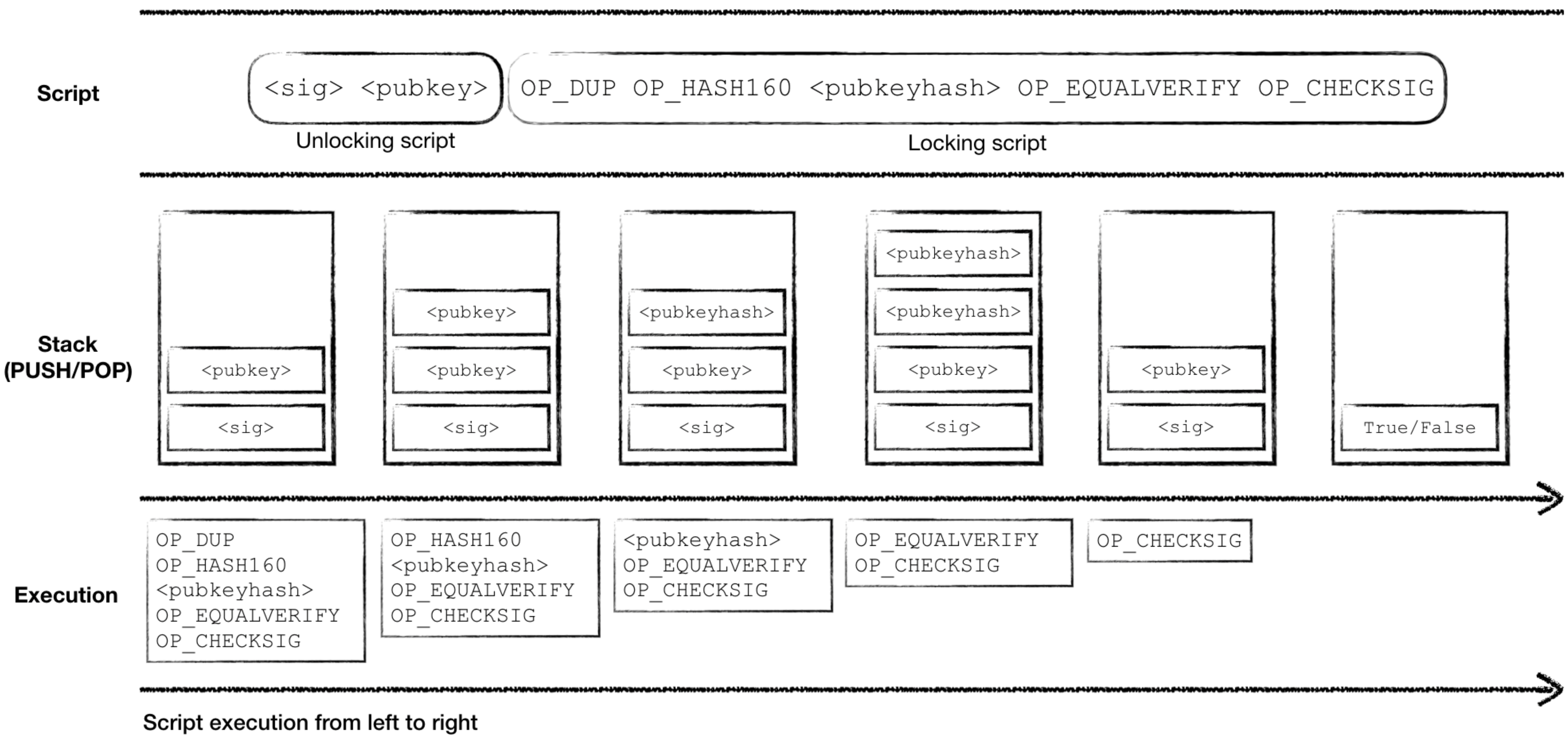
Opcodes

Here are some examples of a few useful opcodes used in the Script language on the Bitcoin blockchain.

Opcode	Description
OP_CHECKSIG	This takes a public key and signature and validates the signature of the hash of the transaction. If it matches, then TRUE is pushed onto the stack; otherwise, FALSE is pushed.
OP_EQUAL	This returns 1 if the inputs are exactly equal; otherwise, 0 is returned.
OP_DUP	This duplicates the top item in the stack.
OP_HASH160	The input is hashed twice, first with SHA-256 and then with RIPEMD-160.
OP_VERIFY	This marks the transaction as invalid if the top stack value is not true.
OP_EQUALVERIFY	This is the same as OP_EQUAL, but it runs OP_VERIFY afterward.
OP_CHECKMULTISIG	This instruction takes the first signature and compares it against each public key until a match is found and repeats this process until all signatures are checked. If all signatures turn out to be valid, then a value of 1 is returned as a result; otherwise, 0 is returned.
OP_HASH256	The input is hashed twice with SHA-256.
OP_MAX	This returns the larger value of two inputs.

P2PKH script execution

P2PKH is the most commonly used transaction type and is used to send transactions to Bitcoin addresses.



Transaction validation

During validation, the following are checked:

- That transaction inputs are previously unspent. This validation step prevents double-spending by verifying that the transaction inputs have not already been spent by someone else.
- That the sum of the transaction outputs is not more than the total sum of the transaction inputs. However, both input and output sums can be the same, or the sum of the input (total value) could be more than the total value of the outputs. This check ensures that no new bitcoins are created out of thin air.
- That that the digital signatures are valid, which ensures that the script is valid.

Blocks

The data structure of a Bitcoin block is shown in the following table:

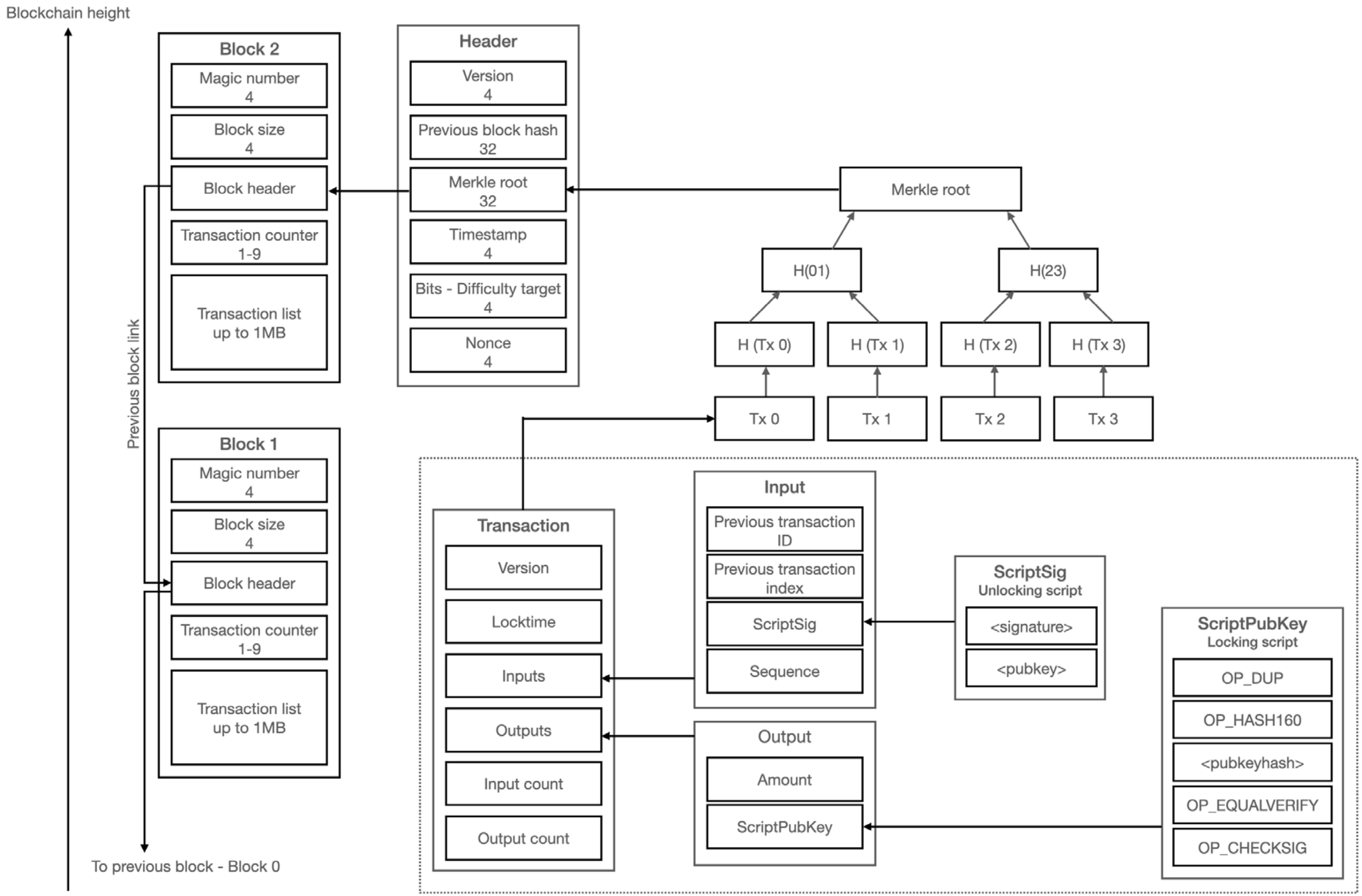
Field	**Size**	**Description**
Block size	4 bytes	The size of the block.
Block header	80 bytes	This includes fields from the block header described in the next section.
Transaction counter	Variable	The field contains the total number of transactions in the block, including the coinbase transaction. Size ranges from 1-9 bytes.
Transactions	Variable	All transactions in the block.

Blocks

The structure of a block header is shown here:

Field	Size	Description
Version	4 bytes	The block version number that dictates the block validation rules to follow.
Previous block's header hash	32 bytes	This is a double SHA256 hash of the previous block's header.
Merkle root hash	32 bytes	This is a double SHA256 hash of the Merkle tree of all transactions included in the block.
Timestamp	4 bytes	This field contains the approximate creation time of the block in Unix-epoch time format. More precisely, this is the time when the miner started hashing the header (the time from the miner's location).
Difficulty target	4 bytes	This is the current difficulty target of the network/block.
Nonce	4 bytes	This is an arbitrary number that miners change repeatedly to produce a hash that is lower than the difficulty target.

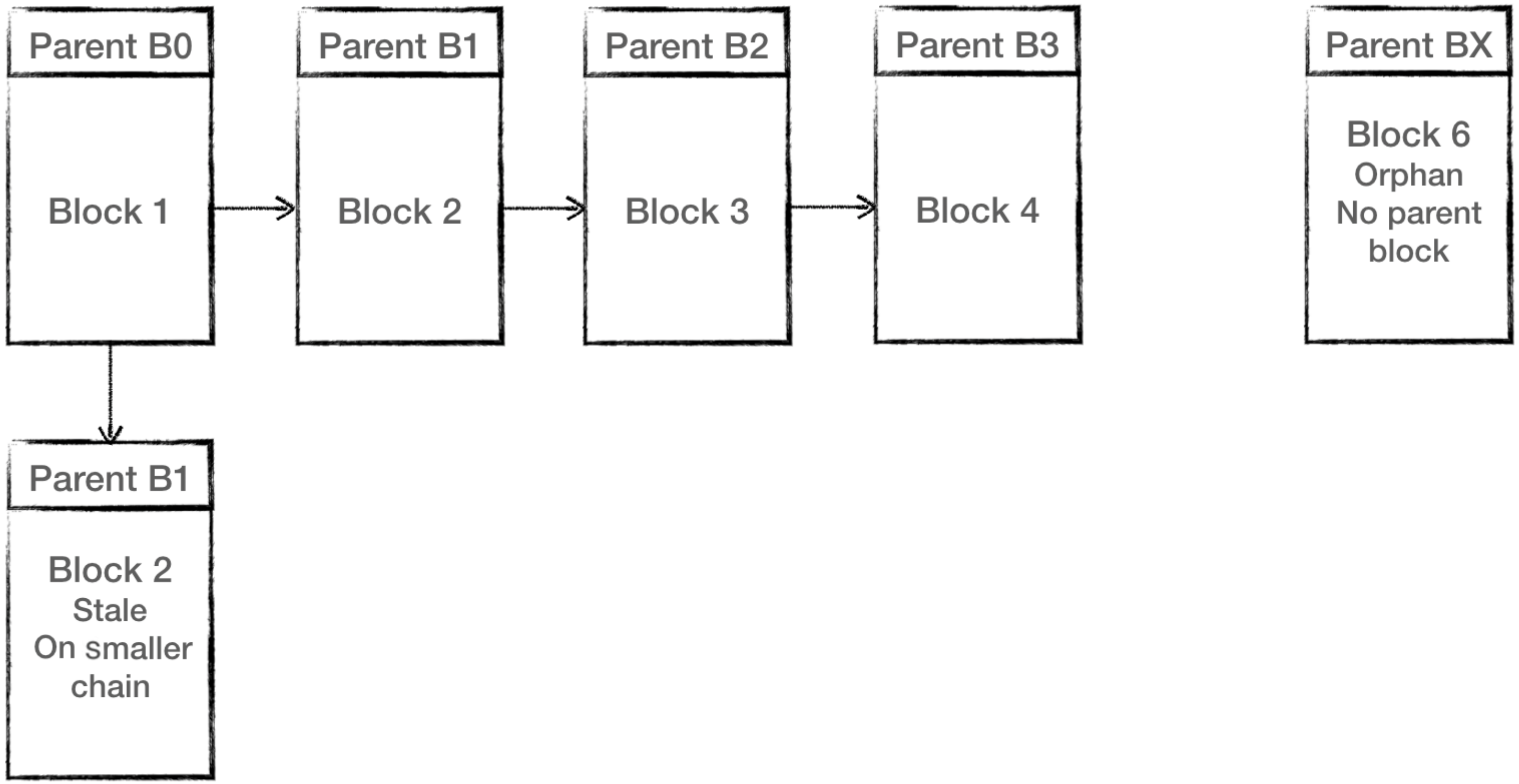
A visualization of the Bitcoin blockchain



Stale and orphan blocks

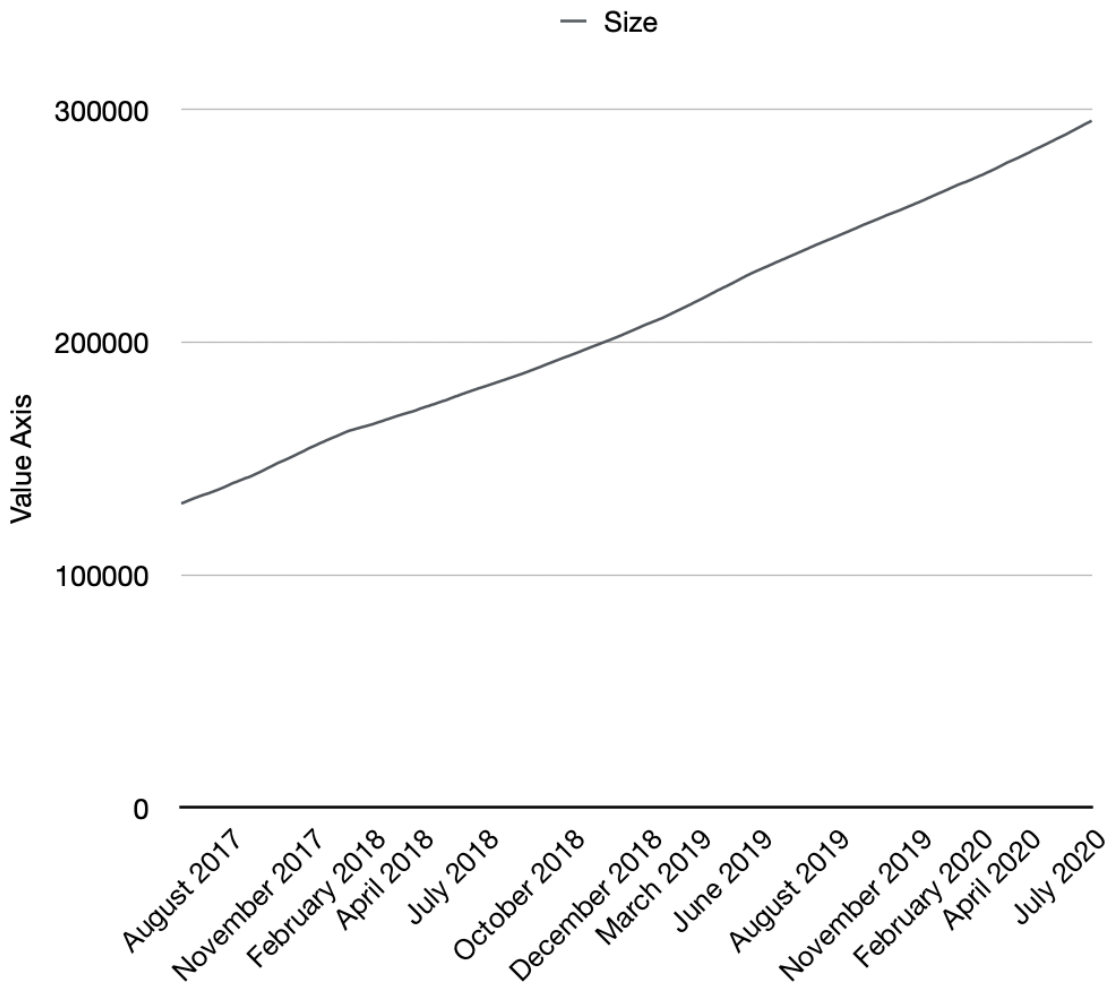
Stale blocks exists on a shorter chain, from which the main chain has progressed

Orphan block's parent blocks are unknown



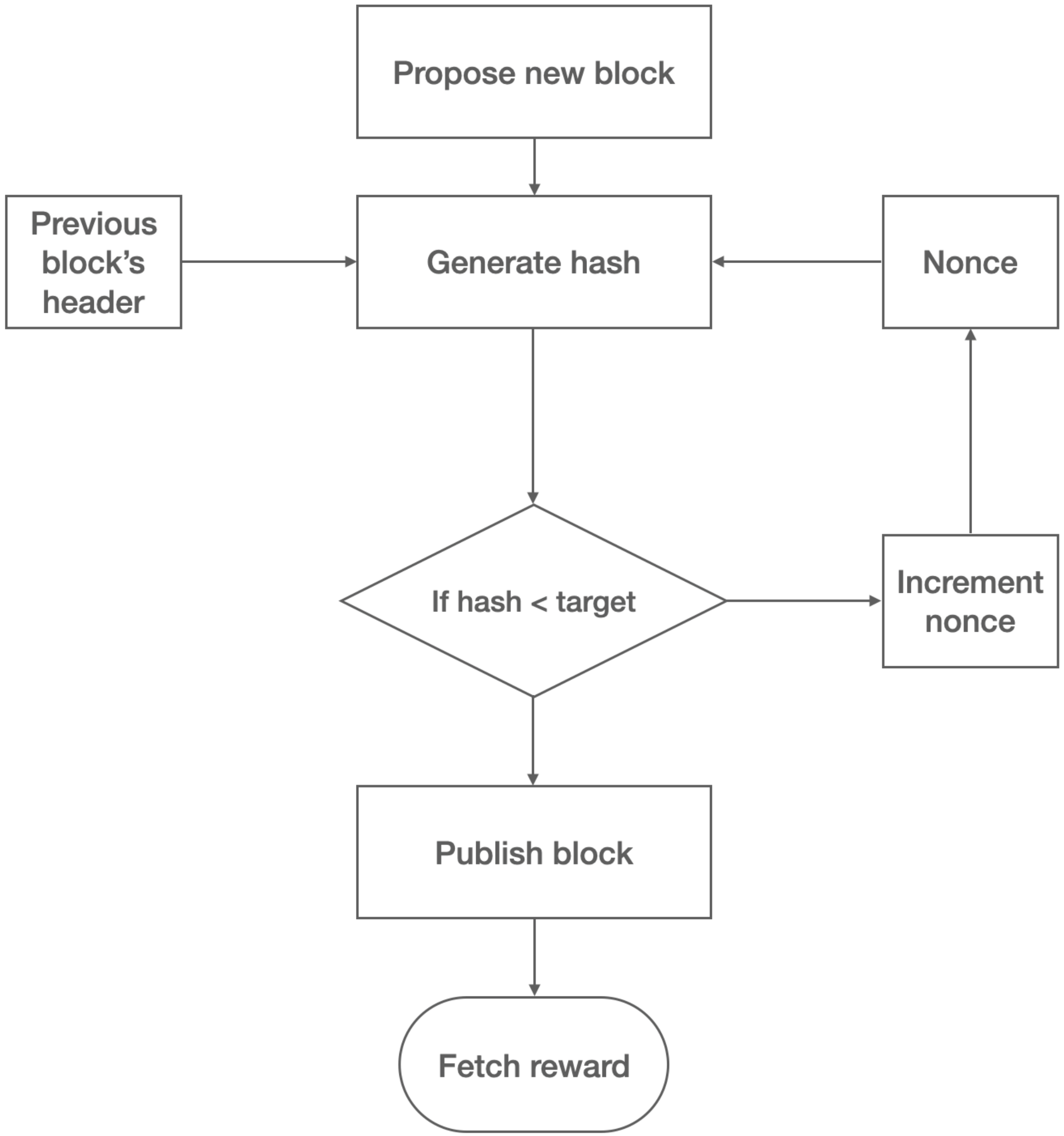
Blockchain size

Bitcoin blockchain size in gigabytes—Aug 2017 to Jul 2020. It is currently approximately 286GB

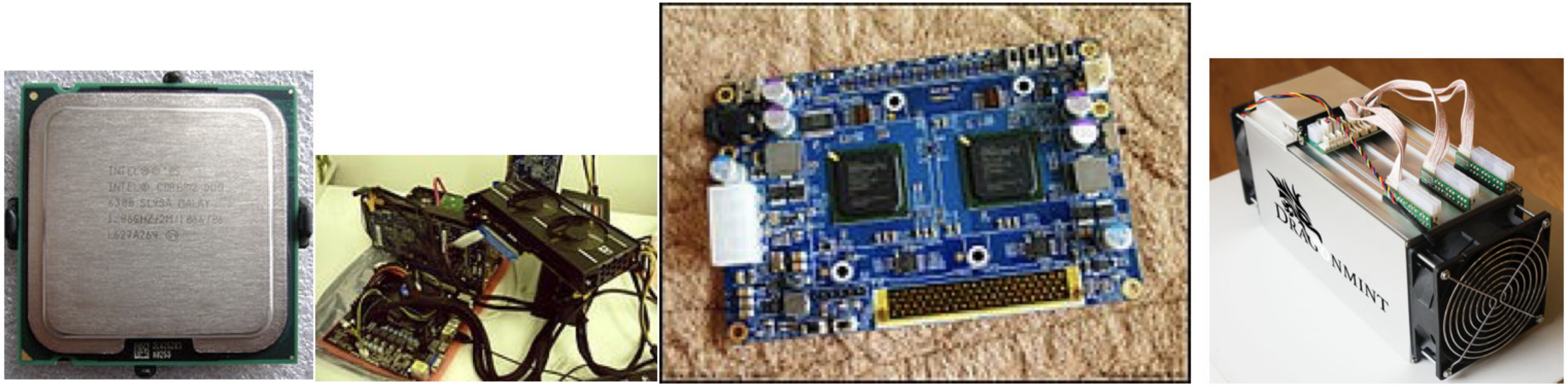


Mining

- Synchronizing with the network
- Transaction validation
- Block validation
- Create a new block
- Perform PoW
- Fetch reward



Mining systems



Four types of mining hardware. From left to right: a CPU, GPU, FPGA, and an ASIC

Exercise

- Install a Bitcoin client, and run experiments on it!

Summary

In this presentation, we covered:

- An introduction to Bitcoin
- How transactions work
- Public and private keys in Bitcoin
- Addresses and their different types
- Transactions and their types and usage
- Blockchain and the various components included in the Bitcoin blockchain